

Application Layer



Chapter 3
CCNA Exploration 1
Part II

SMTP/POP/IMAP Protocols

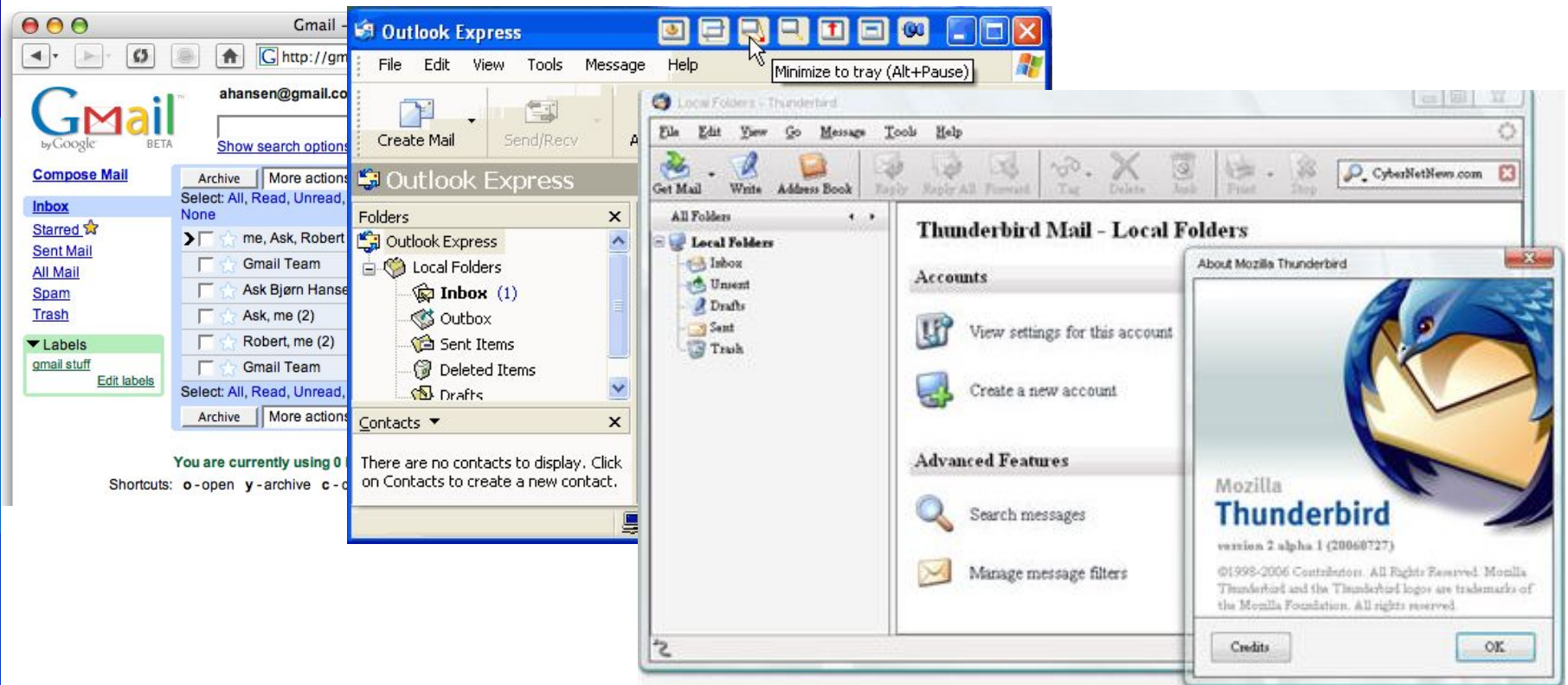
Email Services

Simple Mail Transfer Protocol

Post Office Protocol

Internet Message Access
Protocol

E-Mail Services and SMTP/POP Protocols



□ Revolutionized how people communicate.

Electronic Mail

Three major components:

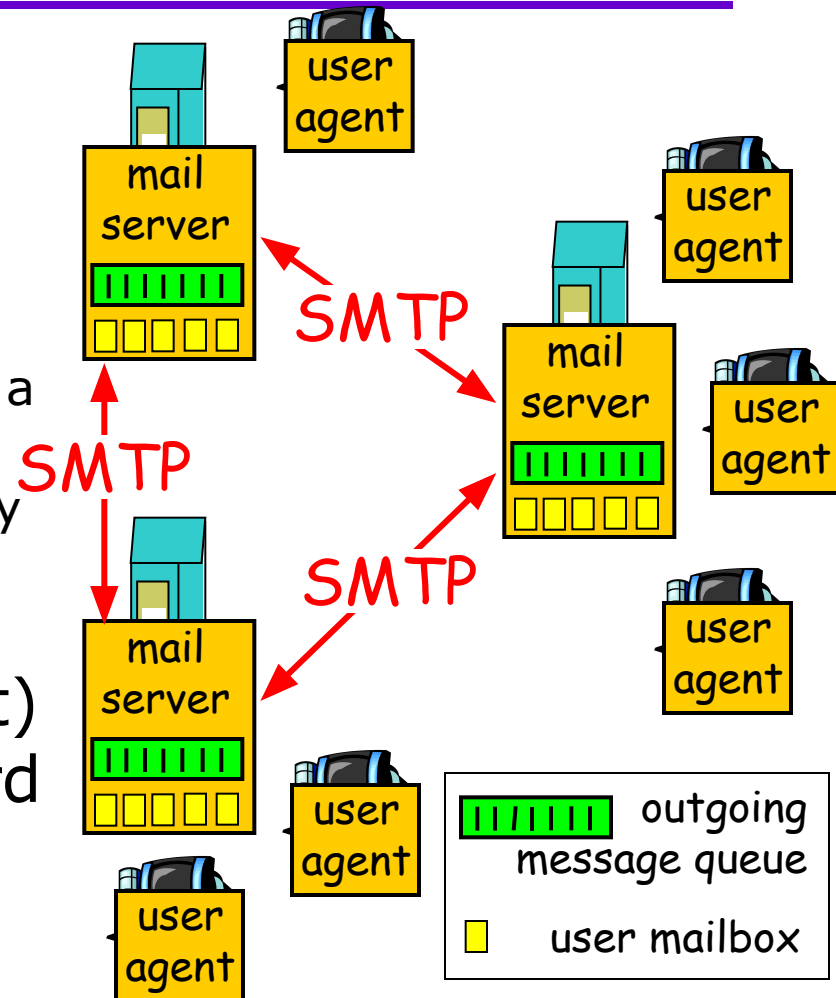
- Mail User agents

- Mail servers

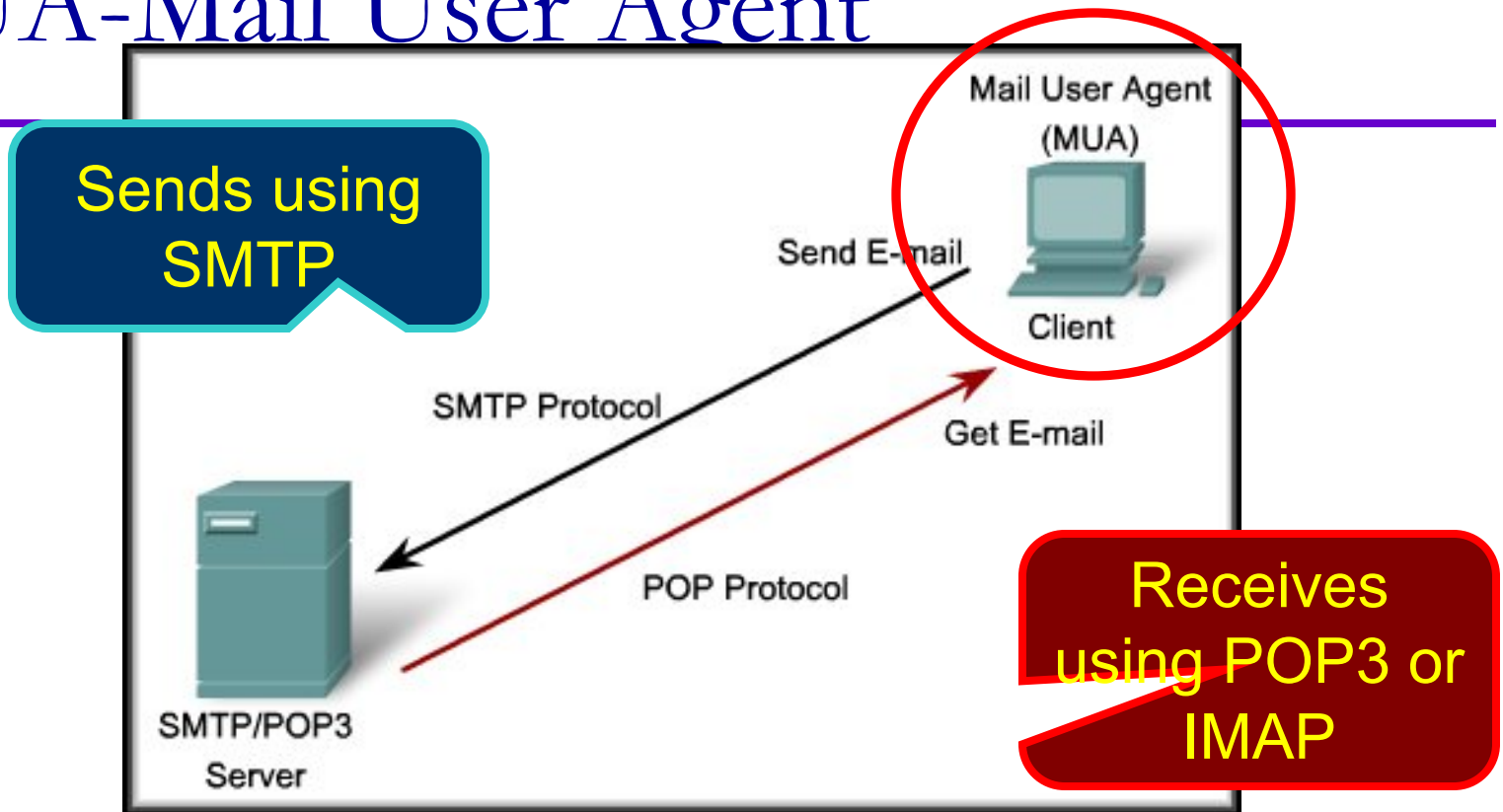
- MDA (Mail Delivery Agent) is a computer software component that is responsible for the delivery of e-mail messages to a local recipient's mailbox.)

- MTA (Mail Transfer Agent) process is used to forward e-mail)

- Protocol: SMTP



MUA-Mail User Agent

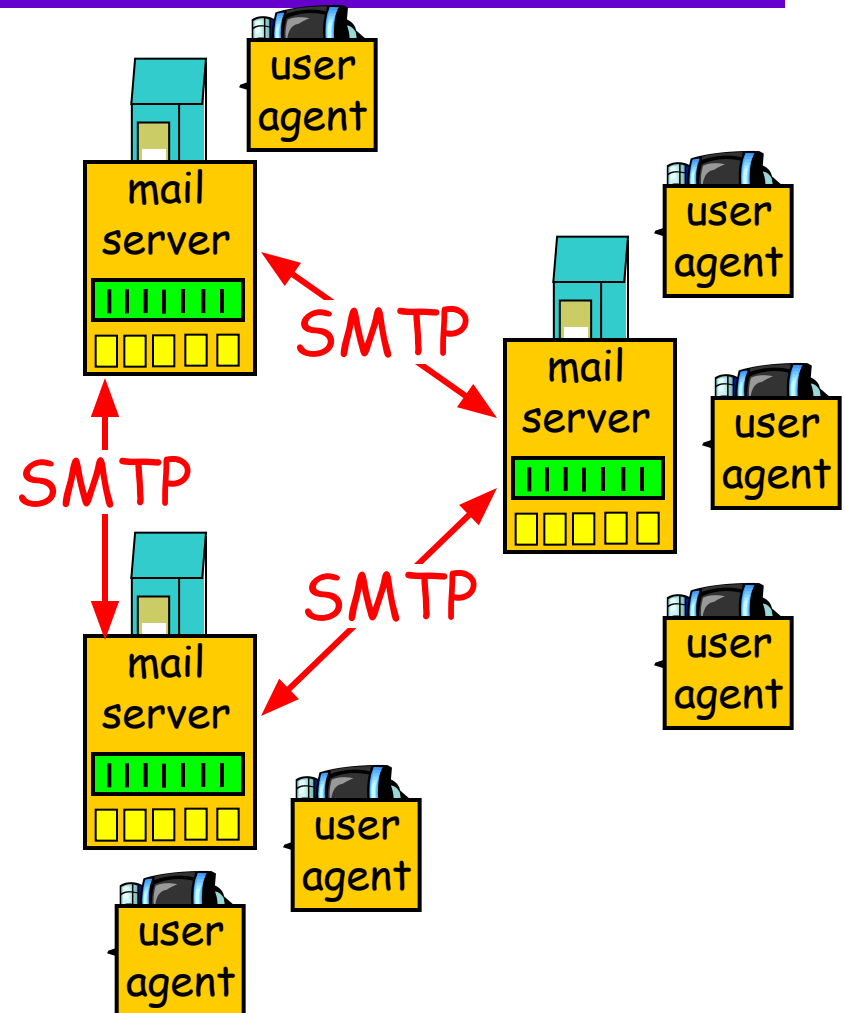


- Mail User Agent (MUA) also known as an email client.
 - Composing, editing, reading mail messages
 - E.g., Eudora, Outlook, Mozilla Thunderbird
 - Outgoing, incoming messages stored on server

Electronic Mail: mail servers

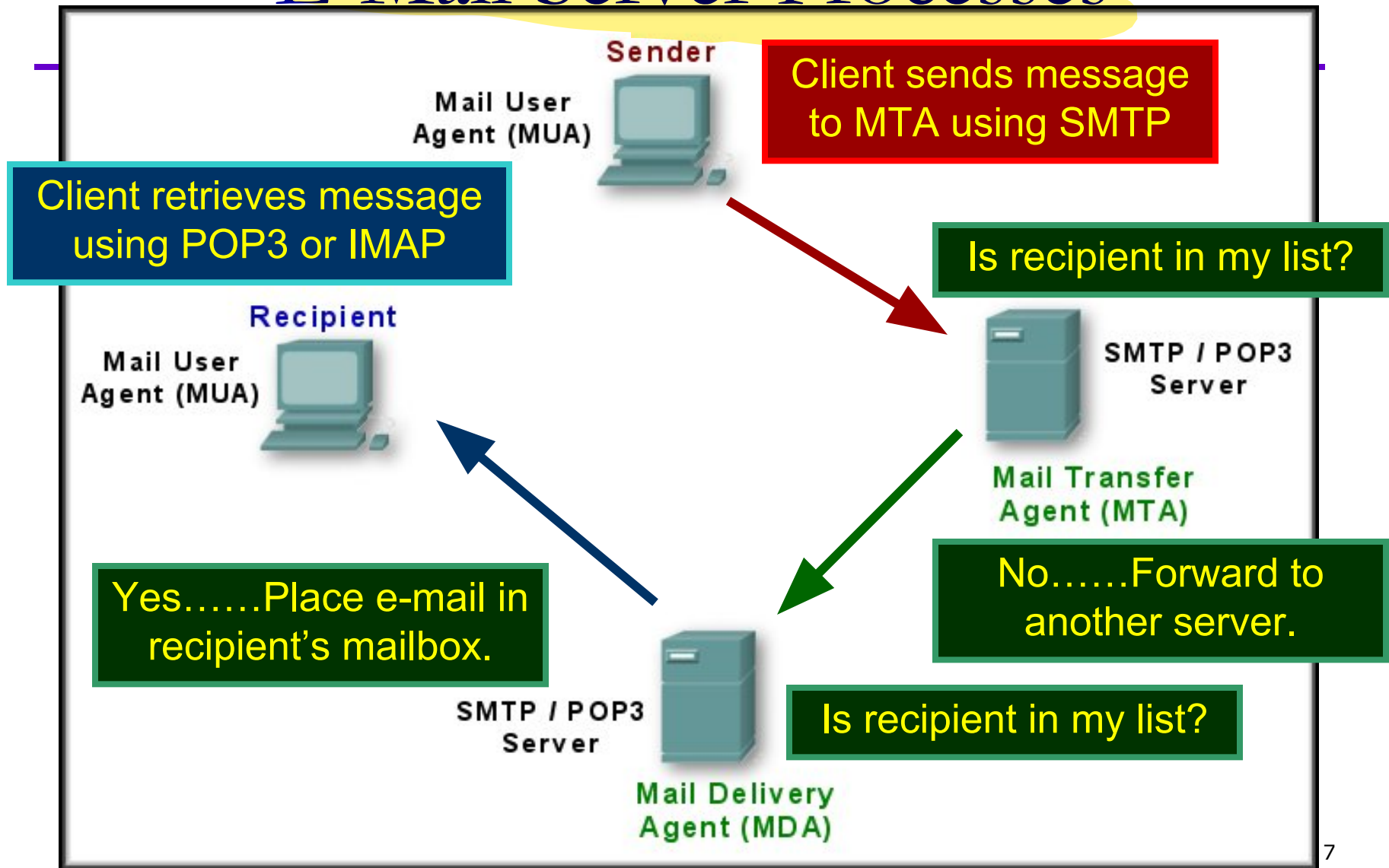
Mail Servers

- **mailbox** contains incoming messages for user
- **message queue** of outgoing (to be sent) mail messages
- **SMTP protocol** between mail servers to send email messages
 - client: sending mail server
 - "server": receiving mail server
 - **PUSH** Protocol



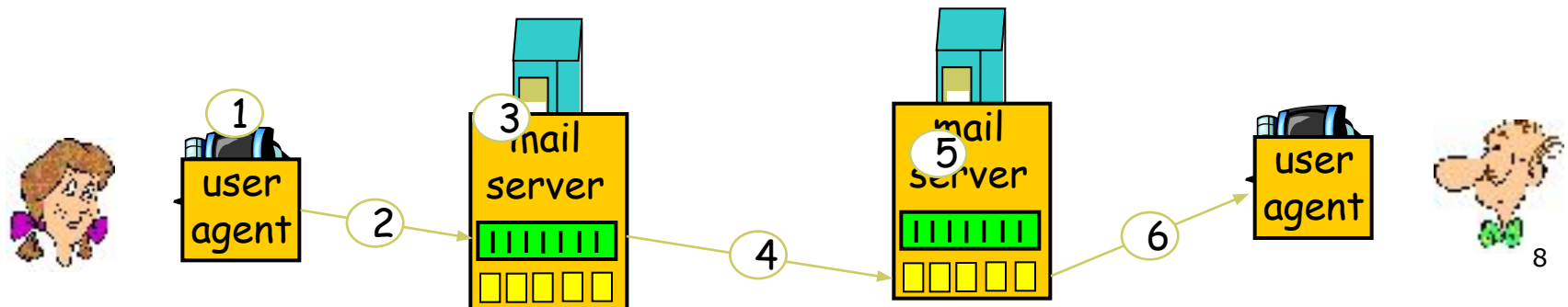
E-Mail Server Processes

→ 15 marks

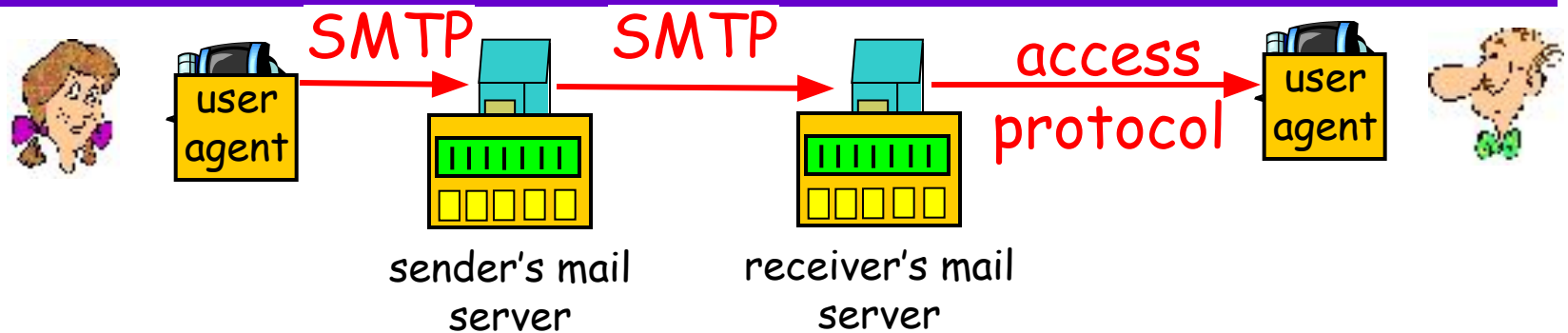


Scenario: Naz sends an email message to Arif

- 1) Naz uses UA to compose message and "to" `arif@someschool.edu`
- 2) Naz's UA sends message to her mail server; message placed in message queue
- 3) Client side of SMTP opens TCP connection with Arif's mail server
- 4) SMTP client sends Naz's message over the TCP connection
- 5) Arif's mail server places the message in Arif's mailbox using POP.
- 6) Arif invokes his user agent to read message



Mail access protocols



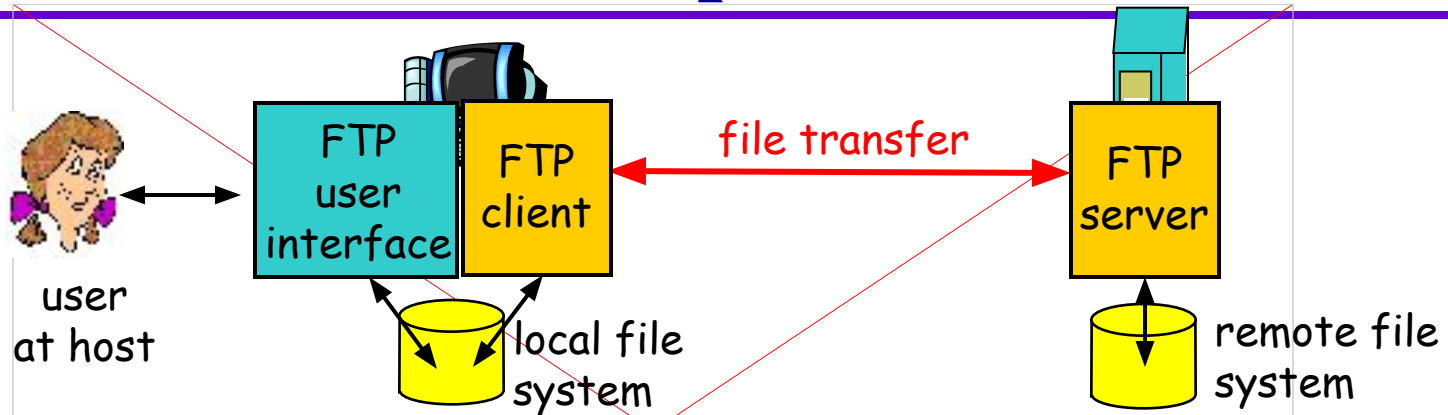
- SMTP: delivery/storage to receiver's server
- Mail access protocol: retrieval from server
 - **POP**: Post Office Protocol [RFC 1939]
 - authorization (agent <-->server) and download
 - **IMAP**: Internet Mail Access Protocol [RFC 1730]
 - more features (more complex)
 - manipulation of stored msgs on server
 - **HTTP**: gmail, Hotmail, Yahoo! Mail, etc.

FTP



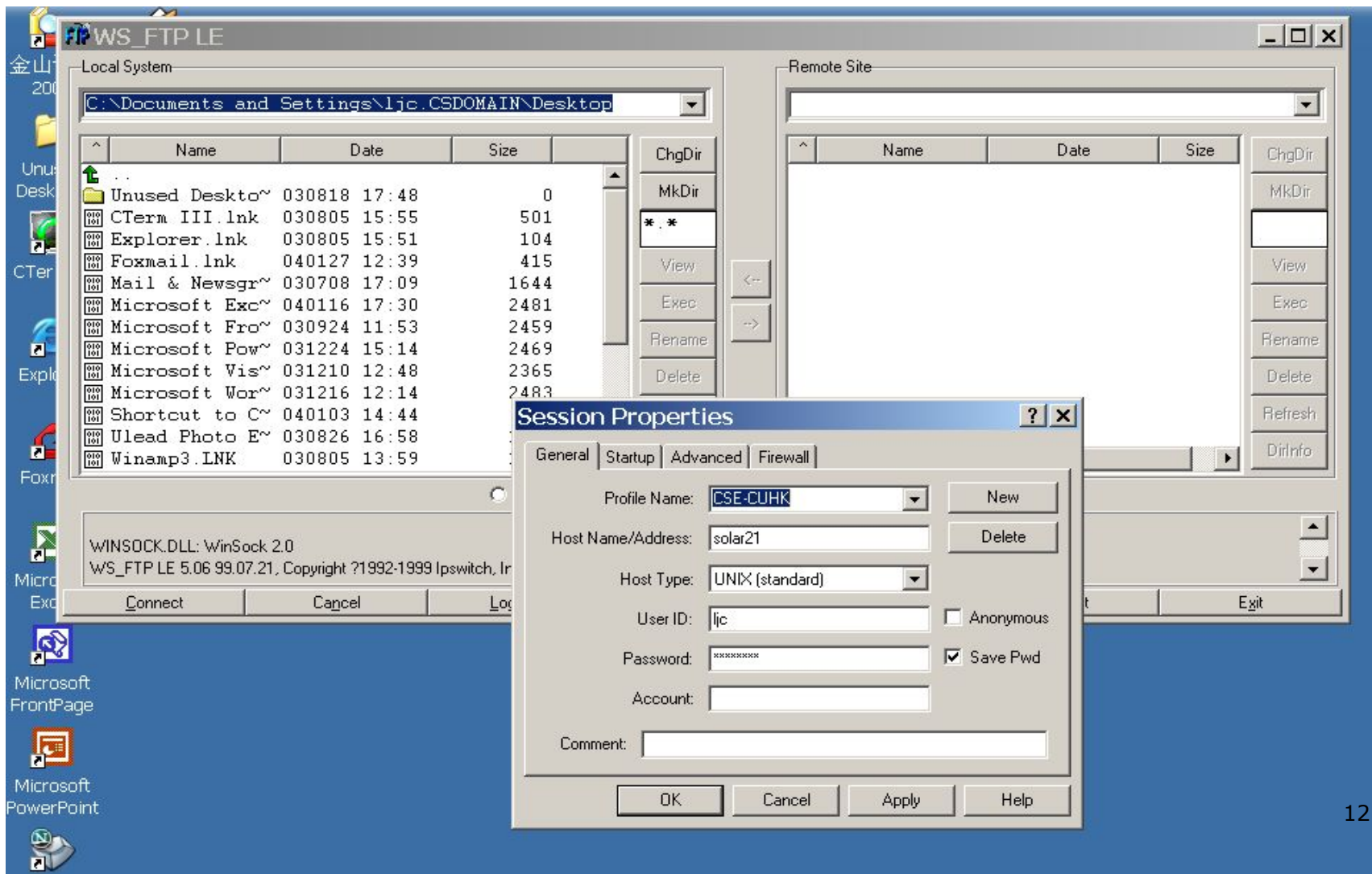
File Transfer Protocol

FTP: the file transfer protocol

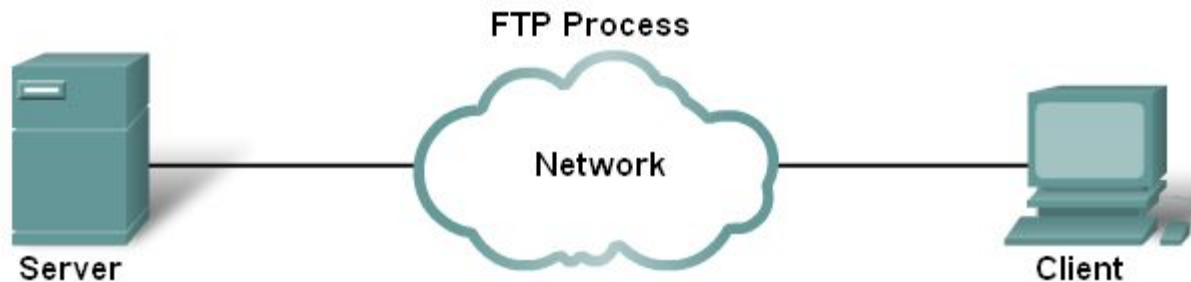


- transfer files between a client and a server.
- client/server model
- ftp: RFC 959
- Requires two connections between the client and the server:
 - one for commands and replies, TCP port 21
 - the other for the actual file transfer, TCP port 20

FTP Client Software



File Transfer Protocol (FTP)



Control Connection:
Client opens first connection to the server for control traffic.



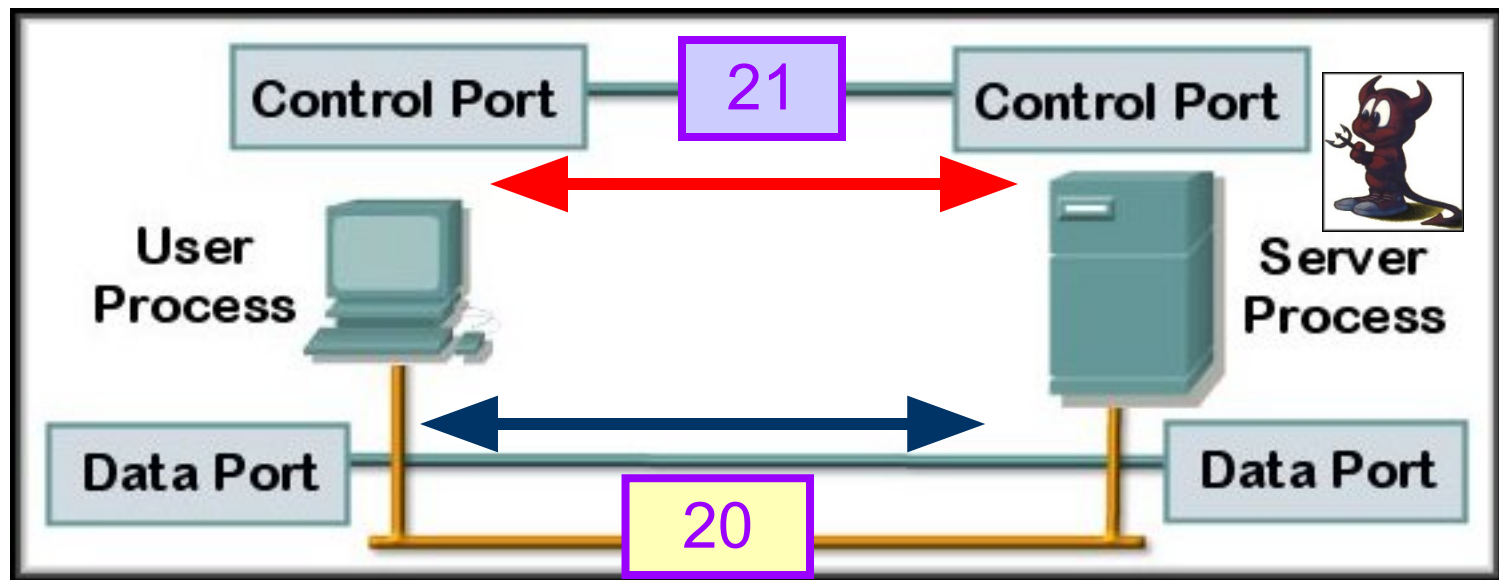
Data Connection:
Client opens second connection for data traffic.



Based on command sent across control connection, data can be downloaded from server or uploaded from client.

File Transfer Protocol (FTP)

Client initiates a TCP control connection on Port 21. Username and password....



For *each file transferred*, TCP opens and closes a Data connection on Port 20.

DHCP

Dynamic Host Configuration
Protocol

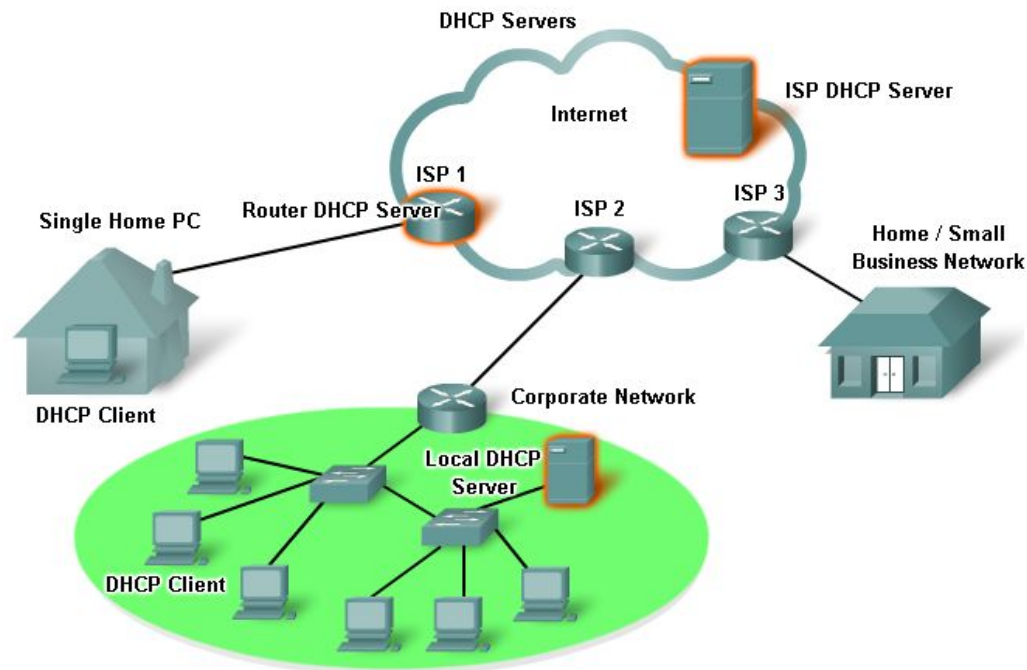
DHCP

- Client /Server
- DHCP service enables devices on a network automatically to obtain IP addresses and other information from a DHCP server.

The screenshot shows the 'General' tab of a network configuration window. It has two tabs: 'General' and 'Alternate Configuration'. The 'General' tab is active. The text inside says: 'You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.' There are two radio button options. The first is 'Obtain an IP address automatically', which is selected. The second is 'Use the following IP address:', which is unselected. Below the second option are three input fields for 'IP address:', 'Subnet mask:', and 'Default gateway:'. There are two more radio button options. The first is 'Obtain DNS server address automatically', which is selected. The second is 'Use the following DNS server addresses:', which is unselected. Below the second option are two input fields for 'Preferred DNS server:' and 'Alternate DNS server:'. At the bottom right of the dialog box is an 'Advanced...' button. At the very bottom are 'OK' and 'Cancel' buttons.

DHCP

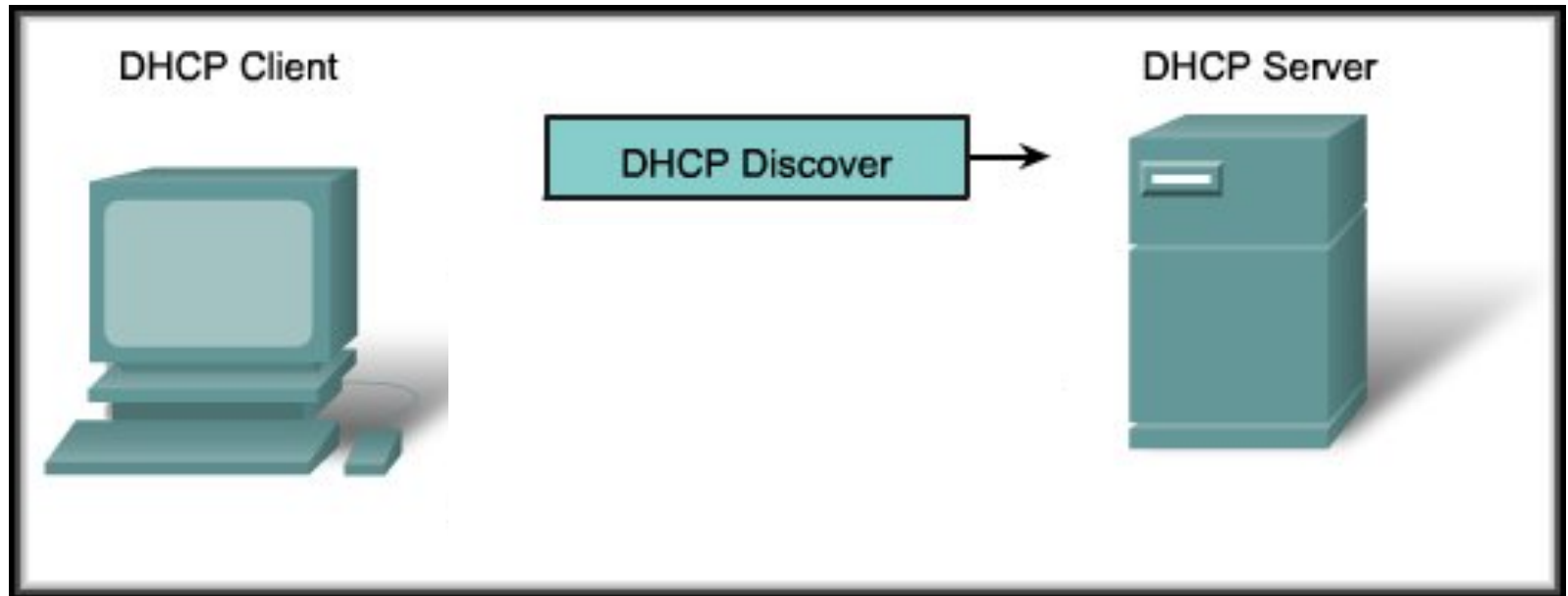
- DHCP servers can be on a LAN, on a router or at an ISP.
- The DHCP server maintains a pool of IP addresses and leases an address to any DHCP-enabled client when the client is powered on.



DHCP Operation



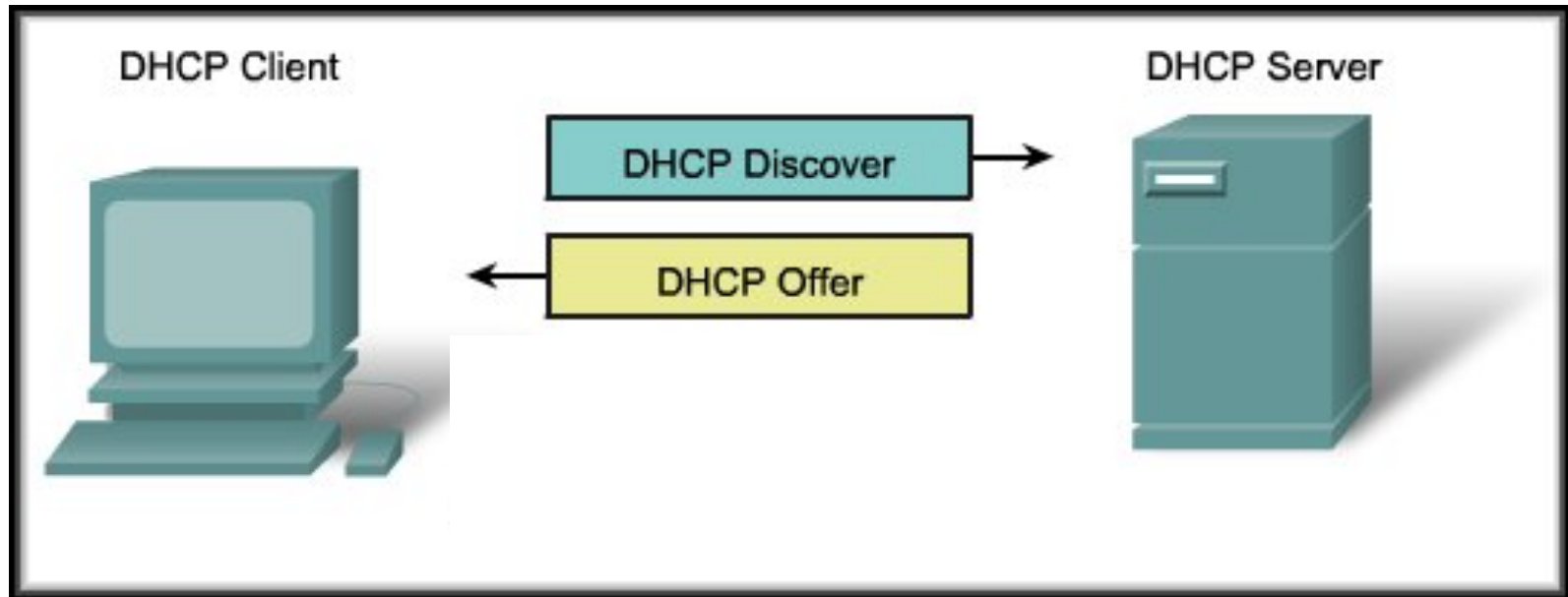
- DHCP is a **four** step process.



1. Client broadcasts a **DHCP Discover** frame to find a DHCP server. There may be more than one available.

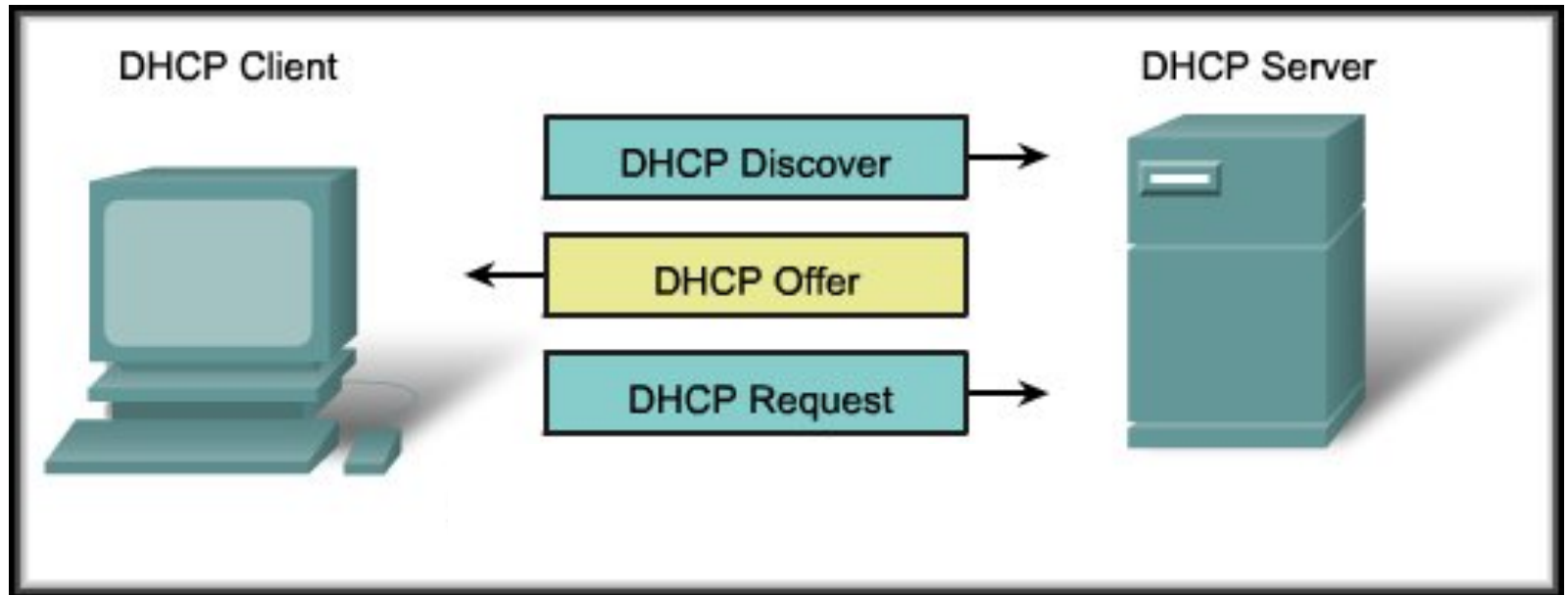
DHCP Operation

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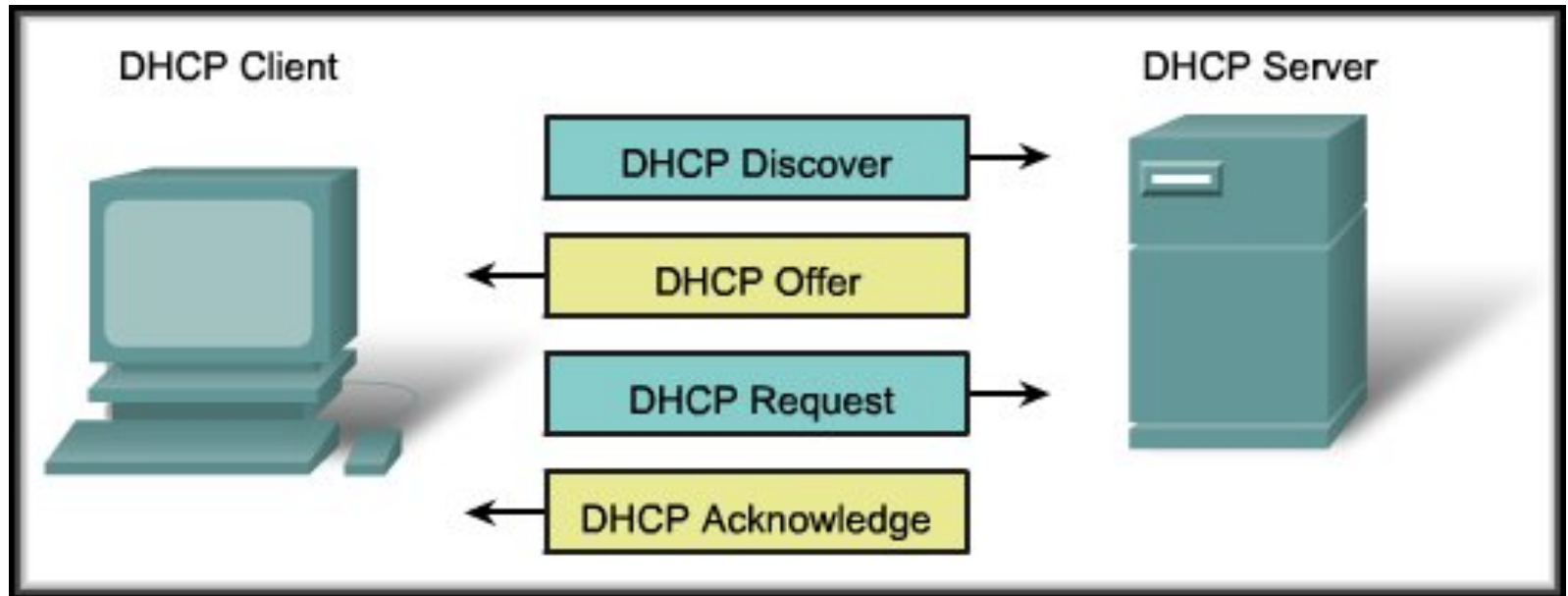
2. A DHCP server responds with a **DHCP Offer** frame containing a lease time, an IP Address, Subnet Mask, and addresses for a Default Gateway and DNS Server.

DHCP Operation



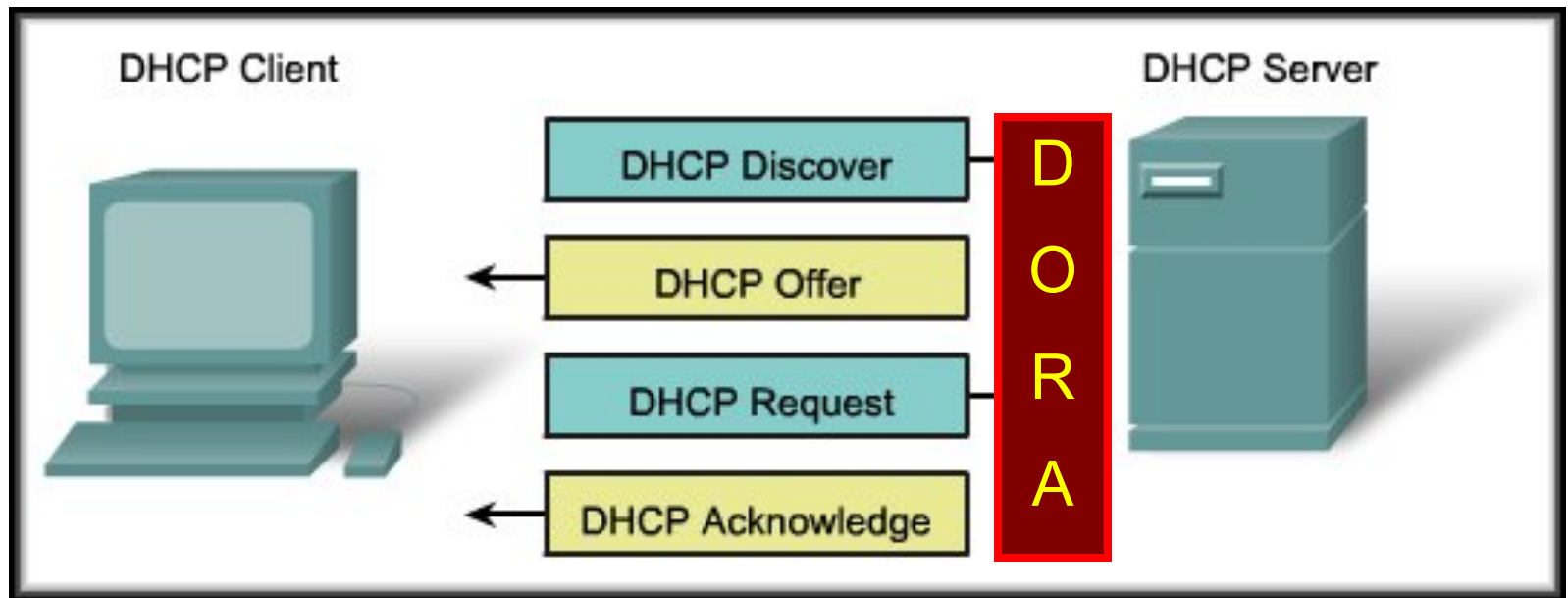
3. The client responds by broadcasting a **DHCP Request** that identifies the server and the lease offer it is accepting.

DHCP Operation



4. If the offer is still valid, the server returns a **DHCP Acknowledgement** and records that information as used. If it is no longer valid, a **DHCP Negative Acknowledgement** is sent and the process begins again.

DHCP Operation



DHCP

- DHCP can pose a security risk because any device connected to the network can receive an address....
- This risk makes physical security an important factor when determining whether to use dynamic or manual addressing.
- **Dynamic** and **static addressing** both have their places in network designs.

DHCP

□ Dynamic Addressing:

- Used for general purpose hosts such as end user devices.



□ Static Addressing:

- Used for network devices such as gateways, switches, servers and printers.



P2P



Peer to Peer

P2P Applications

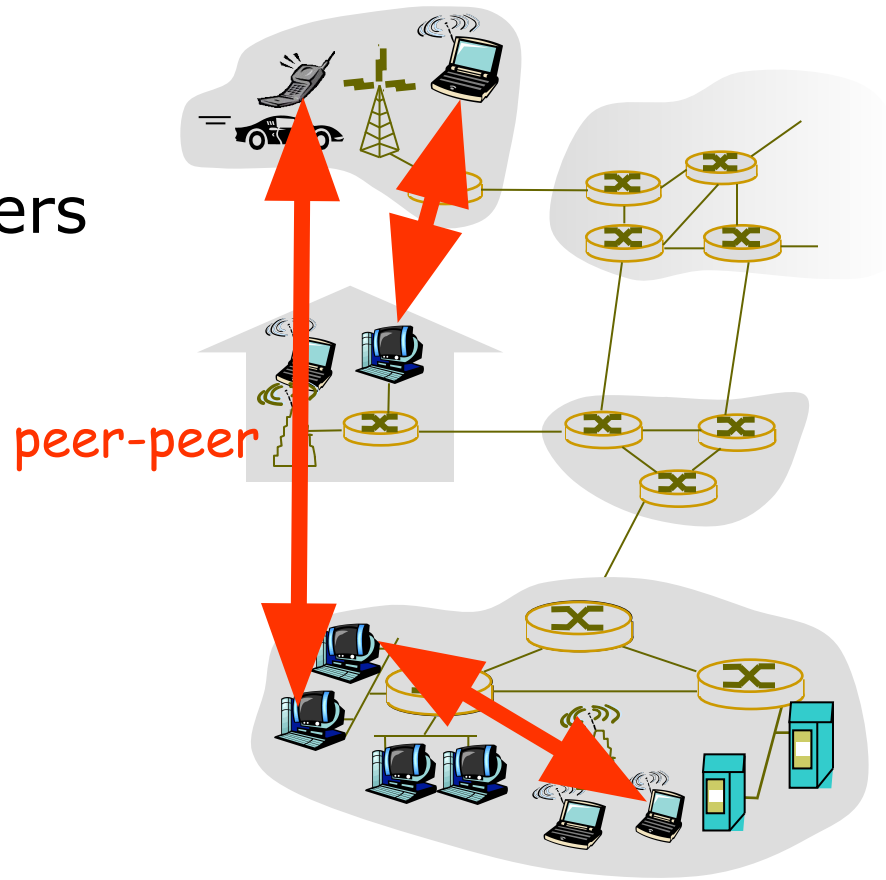


□ Roughly 60% of all upstream traffic and 25% of all downstream traffic on an average day can be attributed to P2P applications in Asia Pacific in 2010.

□ Source :Article from TorrentFreak

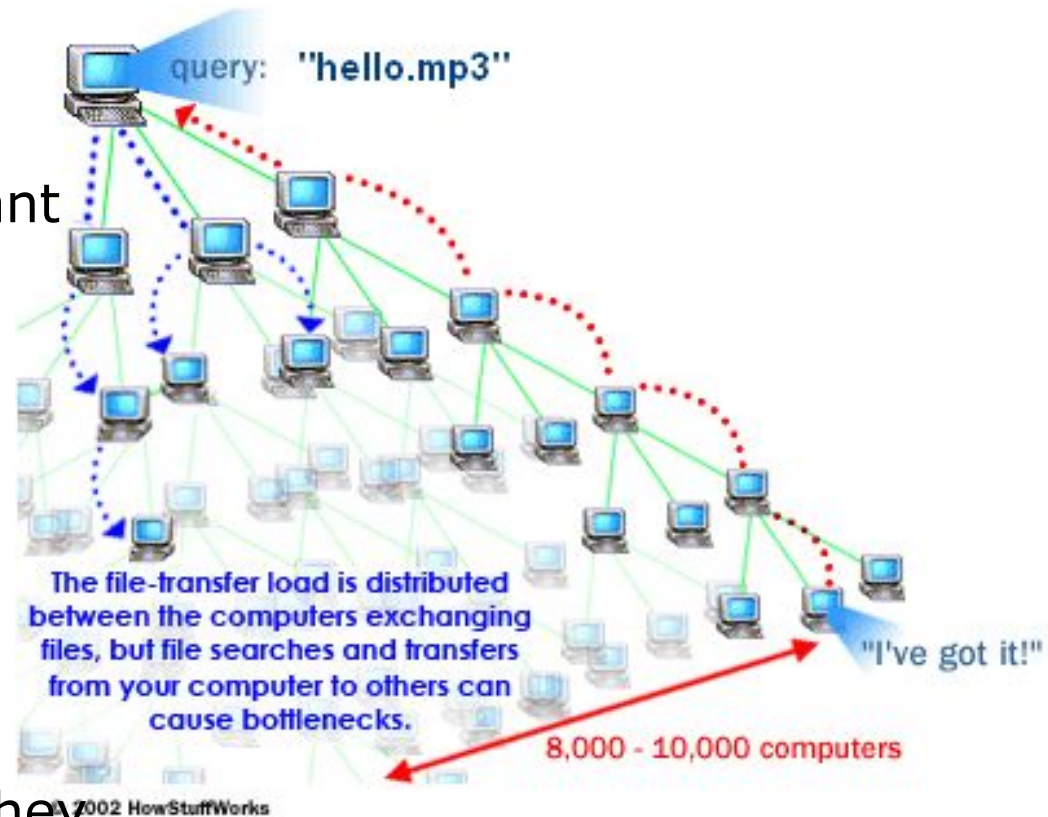
Pure P2P architecture

- No always-on server
- Peers (hosts) act as both clients and servers
- Most popular P2P protocol now is **BitTorrent.**



P2P Applications in general

- ❑ P2P software on your computer sends out a request for the file you want to download.
- ❑ To locate the file, the software queries other computers.
- ❑ When the software finds a computer that has the file the download begins.
- ❑ Others using the P2P software can obtain files they want from your computer's hard drive.

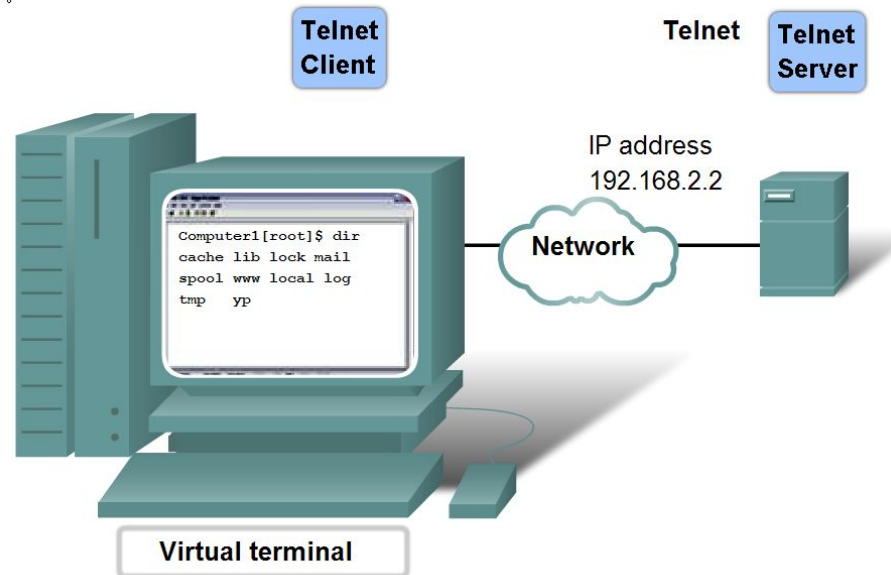


Telnet



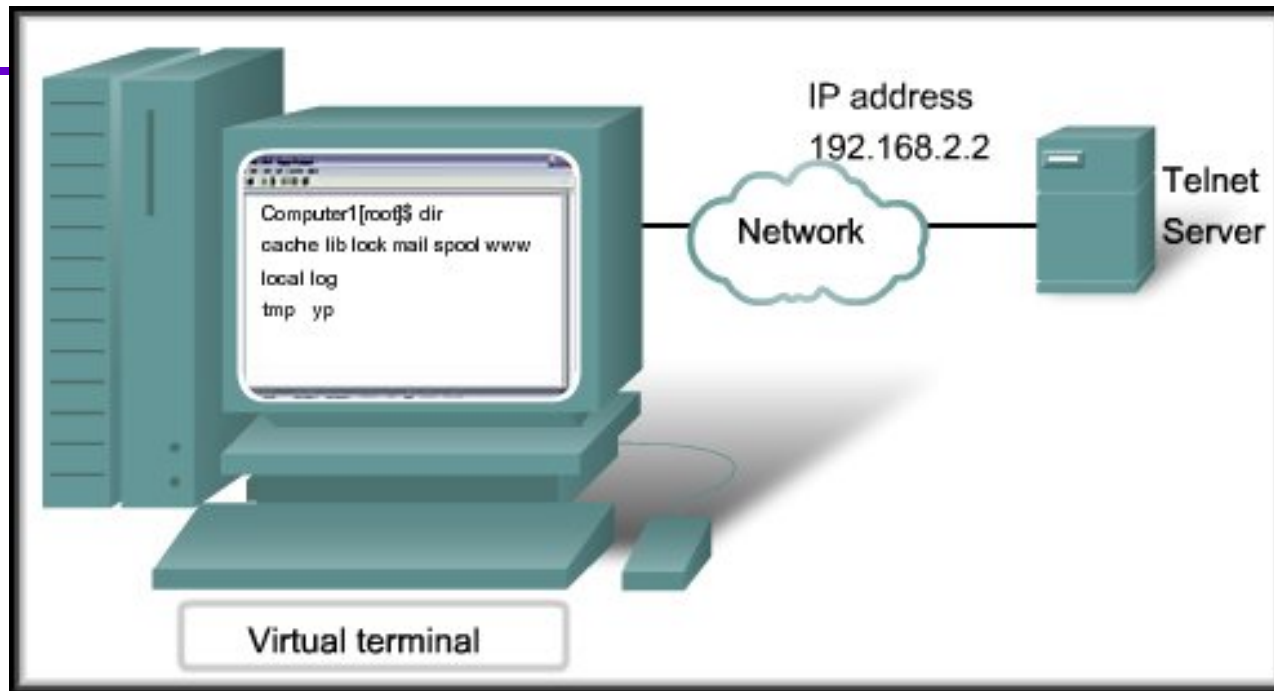
Telnet Services and Protocols

- Once networks were available, people needed a way to **remotely access** the computer systems.



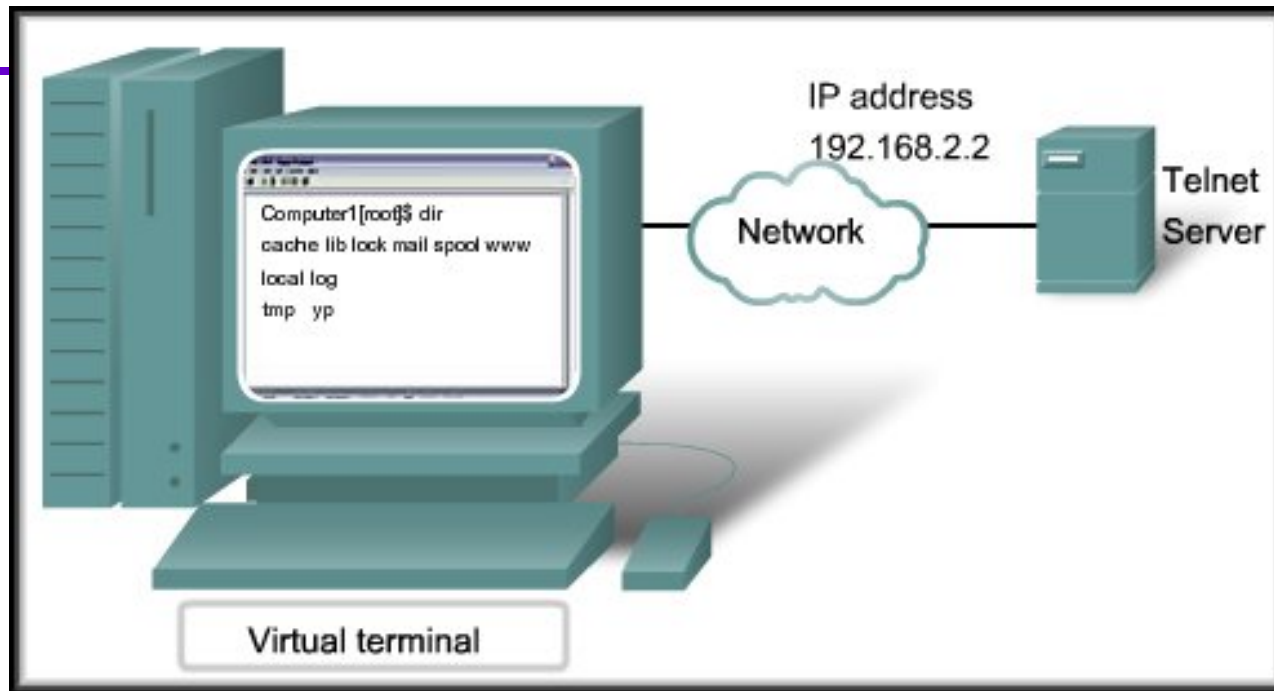
Telnet provides a way to use a computer, connected via the network, to access a network device as if the keyboard and monitor were directly connected to the device.

Telnet Services and Protocol



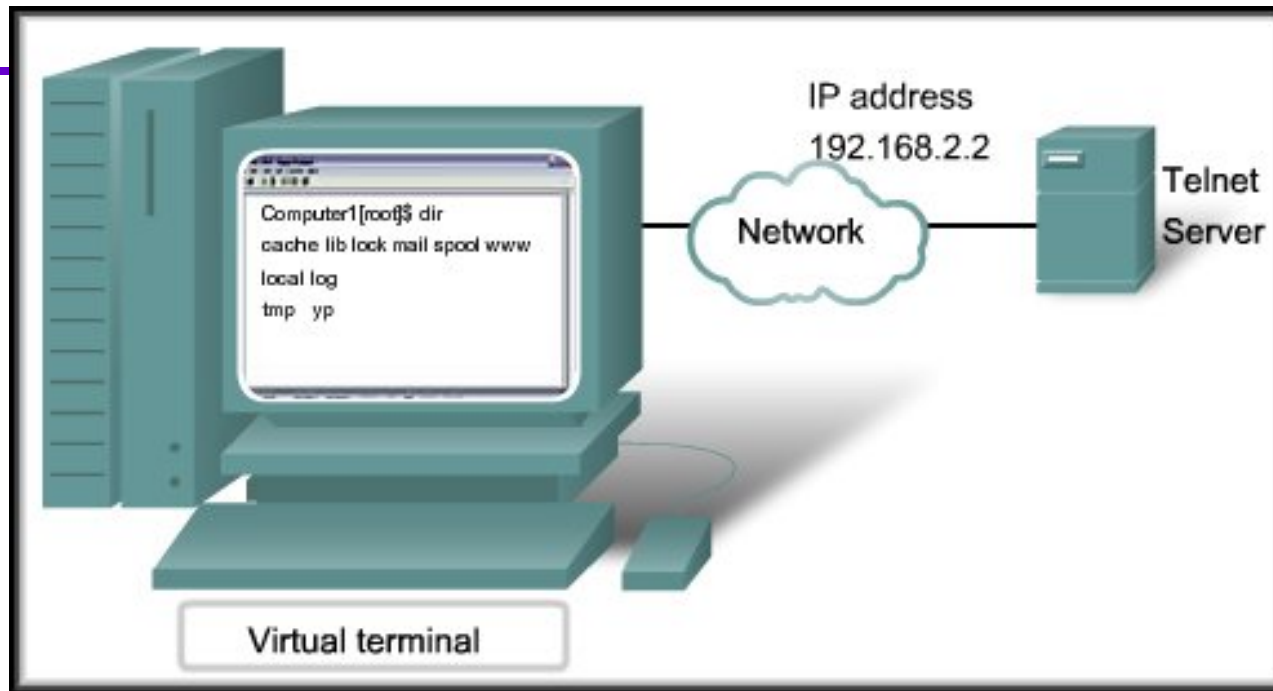
- Allows a user to remotely access another device (host, router, switch).
- A connection using Telnet is called a **Virtual Terminal** (VTY) session or connection.

Telnet Services and Protocol



- Telnet uses software to create a virtual device that offers the same features as a terminal session **command line interface (CLI)**.
- **Telnet Clients:** Putty, Teraterm, HyperTerminal

Telnet Services and Protocol



- Telnet supports user authentication but does not encrypt data (clear text).
- **Secure Shell (SSH)** protocol offers a secure method for server access.
 - Stronger authentication, encrypts data

Telnet Example

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>telnet 192.168.10.222
Trying 192.168.10.222 ...

[Connection to 192.168.10.222 closed by foreign host]
PC>telnet 192.168.10.158
Trying 192.168.10.158 ...

User Access Verification

Password:
HQ>
```

